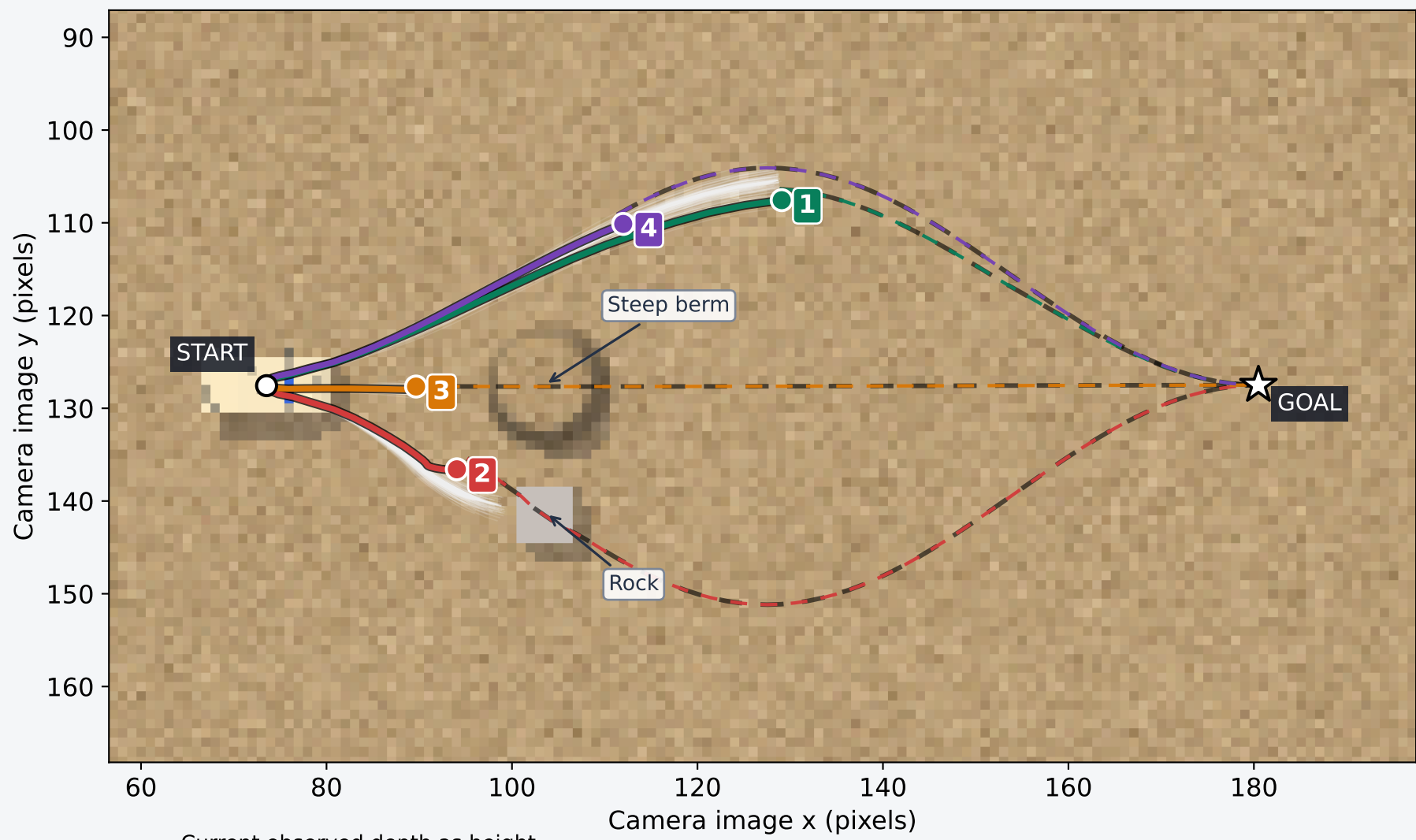


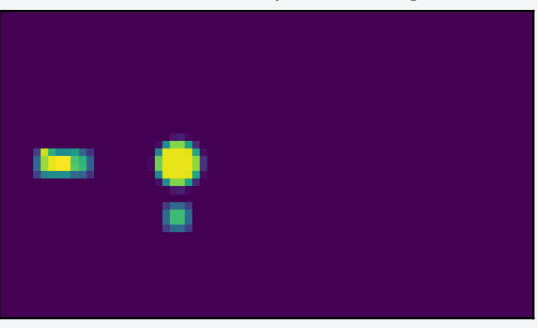
# One scene. Four candidate routes.

Frozen model forecasts before Chrono execution

Current RGB view (cropped) | 592 candidate forecasts



Current observed depth as height



— Solid: learned 4s motion forecast  
- - Dashed: remaining commanded reference (not predicted motion)

Faint curves: all scored candidate forecasts.  
RGB and depth are the same current observation.

Feature names use scene metadata for annotation only;  
they never enter the model or candidate scoring.

## 1 MPPI selected

Predicted goal progress at 4s **17.25 m**  
P(contact <= 4s) <0.001%  
P(confirmed stop <= 4s) <0.001%

## 2 Rock-crossing test

Predicted goal progress at 4s **6.45 m**  
P(contact <= 4s) 0.0103%  
P(confirmed stop <= 4s) 0.0023%

Model predicts poor progress but contact probability is only 0.0103%; collision is a scene hypothesis to validate.

## 3 Terrain-stall test

Predicted goal progress at 4s **5.12 m**  
P(contact <= 4s) 0.0256%  
P(confirmed stop <= 4s) 0.0590%

Model predicts poor progress; confirmed bounded-stop probability by 4 s is only 0.0590%. Later stall may exceed its horizon.

## 4 Slower clear route

Predicted goal progress at 4s **11.97 m**  
P(contact <= 4s) <0.001%  
P(confirmed stop <= 4s) 0.0048%

Less progress is compared at 4 s; this safe 4 m/s reference may still reach the goal later.

XY traces use perspective camera projection with current observed depth; unknown surface height uses z=0. Overlays are not future images.  
Scores are uncalibrated model outputs. Route labels are test hypotheses, not predicted or measured event claims; actual outcomes appear in the videos.